

Onboard Random Network Distillation for Mapless Rover Exploration

Anonymous Authors¹

Abstract

Classical deterministic planners fail in unstructured real-world environments where maps and external instrumentation are unavailable. We present a real-world reinforcement learning pipeline for a physical rover that learns to explore using world feedback alone. Intrinsic reward is derived from a Vision Memory Model that scores how novel each new scene is using Random Network Distillation on live camera frames. We deploy and train directly on hardware with a safety layer constraining motor commands, and propose novel places found — the number of visually distinct scenes accumulated during an evaluation episode — as a practical, instrumentation-free exploration metric.

1. Introduction

Autonomous exploration is usually framed as a mapping problem: estimate free space, detect frontiers, plan to an informative boundary, and update a global map. This framing has led to strong systems when odometry, depth, compute, and calibration are reliable (Yamauchi, 1997; Thrun et al., 2005). It is much less forgiving on small low-cost robots operating with only a monocular camera, a few range sensors, and noisy inertial measurements. In such settings the map is often the least reliable object in the system: it drifts, aliases visually similar corridors, and can fail before learning has an opportunity to improve behavior.

We study a deliberately constrained alternative: mapless exploration from world feedback. A physical rover must learn online, on board, and without external instrumentation, pre-built maps, or simulator pretraining as the source of the final policy. The only feedback available after an action is what the robot itself

observes: camera frames, short-range distance sensors, inertial response, whether the requested motion was executed, and whether safety recovery was required. This places the problem close to reinforcement learning from world feedback (RLxF): the environment itself supplies the evaluative signal, while the designer specifies how to convert raw consequences into a reward.

The central hypothesis is that useful exploration behavior can be obtained without learning raw motor control. Instead, learning should decide where to try to go next at a low temporal frequency, while a deterministic safety-bounded executor handles the immediate actuation details. This separation is important for real hardware. Raw motor reinforcement learning is sample-inefficient, unsafe, and confounds two questions that should be separated: which local direction is worth exploring, and how should the platform realize that command near obstacles? Our current experiments therefore use high-level local motion abstractions, such as a relative heading followed by forward progress until a front-distance threshold is reached.

Intrinsic motivation provides the natural reward source for mapless exploration. Prediction error, episodic novelty, and random network distillation (RND) have all been used to encourage agents to seek states that are new under their learned representation (Pathak et al., 2017; Burda et al., 2019; Savinov et al., 2019). However, transferring this idea to a physical rover introduces two failure modes that are less visible in simulator benchmarks. First, visual novelty alone can reward nuisance changes such as blur, lighting, or seeing the same wall from slightly different distances. Second, a rover can collect positive novelty while physically failing: turning in place, repeatedly touching a wall, or cycling through a corridor. World-feedback reward must therefore be conditioned on executed motion and penalize recovery, zero-progress, and recent revisits.

This paper is an in-progress report on a real-rover RLxF system. The final architecture is still under active development, but the problem formulation and experimental protocol are already fixed enough to be

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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useful. We make three design commitments. First, the final policy is trained and adapted on the real platform rather than obtained through offline behavioral cloning or simulator transfer. Second, any saved human or autonomous trajectories are used for analysis, reward auditing, and diagnostic ablations, not as supervised pretraining data. Third, evaluation emphasizes instrumentation-free quantities available to the robot and a reviewer: physically successful motion, visually novel places found, recovery frequency, and qualitative coverage of a real apartment environment.

Our contribution is the formulation and implementation of a minimal real-world benchmark for low-cost mapless exploration: an onboard RL loop with intrinsic visual memory, safety-gated local actions, and logged world-feedback rewards suitable for offline auditing. The remainder of the paper describes the system components that are independent of the final feature encoder or policy variant.

2. Methodology

2.1. Problem setting

We consider a rover operating in an unknown indoor environment. At time t the robot receives an observation

$$o_t = (I_t, d_t, m_t, a_{t-1}, e_{t-1}), \quad (1)$$

where I_t is the onboard camera image, d_t are short-range distance measurements, m_t denotes inertial and proprioceptive signals, a_{t-1} is the previous high-level action, and e_{t-1} summarizes what the executor actually achieved. The policy selects a low-dimensional local guide a_t , not motor PWM commands. The executor converts a_t into motor commands subject to hard safety constraints. After execution, the robot observes o_{t+1} and receives a scalar reward r_t computed only from onboard world feedback.

The goal is to maximize exploration over a fixed real-time or fixed-step budget. We do not assume a metric map or ground-truth pose. Approximate dead-reckoned pose may be logged for diagnostics, but it is not treated as an oracle. The practical evaluation question is: did the rover visit more distinct places while avoiding repeated unsafe or zero-progress behavior?

2.2. Local action abstraction

The policy acts at a slower rate than the motor controller. In the current strongest prototype, the action is a scalar relative heading

$$a_t \in [-1, 1], \quad \theta_t = \theta_{\max} a_t, \quad (2)$$

with $\theta_{\max} = 75^\circ$. The executor first turns by θ_t relative to the current body frame, then drives forward until the front range sensor reaches a safety threshold or a maximum drive duration expires. This action is intentionally local and egocentric. It does not require global localization, and it biases exploration toward meaningful forward progress rather than short dithering motions.

Other local guide parameterizations, such as short (θ, ρ) targets or curvature-horizon-speed commands, fit the same framework. The important constraint is that the learned policy proposes a motion intent, while the executor remains responsible for safety, clipping, and recovery.

2.3. Onboard novelty memory

The exploration reward is built around an online visual memory model. A feature encoder maps the camera frame and recent action context into an embedding z_t . The memory bank stores representative embeddings of previously visited experience. Novelty is estimated by distance to the nearest memory item and/or by prediction error from an RND-style target-predictor pair (Burda et al., 2019). A simplified novelty term is

$$n_t = \alpha \text{dist}(z_t, \mathcal{M}_{t-1}) + \beta \|f_{\text{pred}}(I_t) - f_{\text{target}}(I_t)\|_2^2, \quad (3)$$

where $\mathcal{M}_{t-1}$ is the current memory bank and f_{target} is fixed. If z_t is farther than a threshold from existing representatives, a new memory item is added or updated slowly.

The exact encoder is not fixed in this draft. We have tested lightweight convolutional features, recurrent path-conditioned memory, and frozen visual backbones. The paper’s final claims will depend on the version that survives real-world ablations. The methodology, however, does not rely on a particular encoder: any onboard representation can be inserted as long as it produces online novelty estimates from the robot’s own observations.

2.4. World-feedback reward

Naively rewarding novelty is not enough on hardware. A robot can observe new pixels while failing to make useful progress. We therefore compute reward from both perceptual novelty and execution feedback:

$$r_t = w_n \mathcal{K}[\text{motion}_t] n_t + w_c \mathcal{K}[\text{new cluster}_t] + w_d \Delta s_t + w_s \mathcal{K}[\text{safe motion}] \quad (4)$$

$$- \lambda_r R_t - \lambda_0 Z_t - \lambda_l L_t - \lambda_o O_t. \quad (5)$$

Here Δs_t is estimated executed forward distance, R_t denotes recovery or reverse events, Z_t denotes zero-

forward or already-at-front outcomes, L_t penalizes recent local revisits, and O_t penalizes near-obstacle operation. The novelty term is motion-gated, so visual changes collected while stuck or only turning in place are not treated as successful exploration.

This reward is best viewed as an auditable interface between the physical world and the learning algorithm. Each term is logged separately, enabling post-run comparison between scalar reward and human visual judgment. In practice this has been essential: early rewards over-valued safe-looking corridor loops, while later loop-aware terms better separated broad apartment coverage from repeated wall contacts.

2.5. Training and data usage

We use an off-policy continuous-control learner, currently Soft Actor-Critic (Haarnoja et al., 2018), because it can update from the short and noisy streams available during real deployments. The replay buffer is populated online from the rover’s own experience. Safety checks run outside the policy and can override or clip execution; the policy receives the resulting feedback so that failed intents are not recorded as successful motion.

Saved runs are used in a restricted way. Manual driving and autonomous trajectories provide camera frames, sensor traces, reward terms, and human annotations such as “covered all rooms” or “looped in corridor.” These data are valuable for debugging reward functions and evaluating candidate visual memories offline. They are not used to pretrain the final exploration policy, preserving the RLxF setting in which policy improvement comes from real online interaction with the world.

2.6. Evaluation protocol

Our planned evaluation uses multiple short autonomous episodes in the same apartment environment. Primary measurements are: number of visually distinct places discovered, fraction of steps with successful forward execution, recovery count, zero-progress count, approximate path spread from dead reckoning, and qualitative frame review. We additionally keep representative autonomous runs as frozen diagnostic datasets for reward-alignment tests. One such run contains 200 autonomous steps that reached all five rooms but included many corridor loops and wall contacts; it is a useful example because a good metric should recognize both its high coverage and its local-control failures.

3. Ablations

Draft notes from current rover runs. These are working notes, not final claims. The strongest current empirical observation is that the action interface matters more than the particular visual-memory variant: short vector or (θ, ρ) local targets tended to produce timid motion, while a scalar relative-heading action followed by forward execution until a front-range threshold produced the first run that reached all five rooms. This suggests an ablation between fixed-distance local targets and world-bounded execution, where the policy chooses direction but onboard feedback determines the realized travel horizon. A second candidate ablation is requested-action memory versus executed-action memory: using what the rover actually did after safety clipping/recovery appears more faithful than storing only the intended command. The dual-memory patch, separating path/action-conditioned memory from raw visual memory, is worth testing but should not yet be claimed as a final architecture; current visual clustering still needs stronger evidence.

4. Conclusion & Future Work

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